

performance analysis of super market

The background image shows the interior of a supermarket. The ceiling is a prominent feature, with a series of parallel, wavy wooden slats that are illuminated from below, creating a warm, golden glow. The floor is a light-colored, polished tile. On the left, there are tall shelves stocked with various packaged goods, including instant noodle cups. In the center, there are several promotional displays, including a large one for 'Artisan of Genuine Pastries' and another for 'Doritos'. To the right, there are refrigerated display cases filled with fresh produce, including green leafy vegetables. The overall atmosphere is clean, modern, and well-lit.

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➤ Project Overview

Conducted comprehensive Analysis on supermarket sales data using SQL , identifying key drivers of revenue across diverse product lines.

☰ SAMPLE DATA

Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.5775	5	0.45	-383.031
Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies	Storage	22.368	2	0.2	2.5164
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Furnishings	48.86	7	0	14.1694
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Art	7.28	4	0	1.9656
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	907.152	6	0.2	90.7152
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Binders	18.504	3	0.2	5.7825
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Appliances	114.9	5	0	34.47
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Tables	1706.184	9	0.2	85.3092
Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	911.424	4	0.2	68.3568
Standard Class	Consumer	United States	Concord	North Carolina	28027	South	Office Supplies	Paper	15.552	3	0.2	5.4432
Standard Class	Consumer	United States	Seattle	Washington	98103	West	Office Supplies	Binders	407.976	3	0.2	132.5922
Standard Class	Home Office	United States	Fort Worth	Texas	76106	Central	Office Supplies	Appliances	68.81	5	0.8	-123.858
Standard Class	Home Office	United States	Fort Worth	Texas	76106	Central	Office Supplies	Binders	2.544	3	0.8	-3.816
Standard Class	Consumer	United States	Madison	Wisconsin	53711	Central	Office Supplies	Storage	665.88	6	0	13.3176
Standard Class	Corporate	United States	Fremont	Nebraska	68025	Central	Office Supplies	Art	19.46	7	0	5.0596
Standard Class	Corporate	United States	Fremont	Nebraska	68025	Central	Office Supplies	Appliances	60.34	7	0	15.6884
Standard Class	Consumer	United States	Orem	Utah	84057	West	Furniture	Tables	1044.63	3	0	240.2649
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furniture	Bookcases	3083.43	7	0.5	-1665.0522
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Office Supplies	Binders	9.618	2	0.7	-7.0532
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furniture	Furnishings	124.2	3	0.2	15.525
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Office Supplies	Envelopes	3.264	2	0.2	1.1016
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Office Supplies	Art	86.304	6	0.2	9.7092
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Office Supplies	Binders	6.858	6	0.7	-5.715
Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Office Supplies	Art	15.76	2	0.2	3.546
Standard Class	Home Office	United States	Houston	Texas	77041	Central	Office Supplies	Envelopes	113.328	9	0.2	35.415
Standard Class	Home Office	United States	Houston	Texas	77041	Central	Furniture	Bookcases	532.3992	3	0.32	-46.9764

»» ATTRIBUTE'S OF SUPER MARKET

- ➔ **Ship-Mode** – Method used to ship the order (e.g., Standard, First Class).
- ➔ **Segment** – Type of customer (Consumer, Corporate, or Home Office).
- ➔ **Country** – Country where the sale occurred.
- ➔ **City** – City where the order was placed.
- ➔ **State** – State or province of the customer.
- ➔ **Postal-Code** – Postal or ZIP code of the shipping address.
- ➔ **Region** – Geographical region (East, West, Central, South, etc.).
- ➔ **Category** – Main product category (Furniture, Office Supplies, Technology).
- ➔ **Subcategory** – More specific product type (e.g., Chairs, Phones, Binders).
- ➔ **Sales** – Total revenue from the sale.
- ➔ **Quantity** – Number of units sold.
- ➔ **Discount** – Discount applied to the sale.
- ➔ **Profit** – Profit earned after subtracting costs and discounts.

»» SQL Queries for Supermarket Sales Data

- write a query to display the ship mode, sub-category, profit, sales, raw cost of the product ?

QUERY :-

```
SELECT `ship mode`, `sub-category`, profit, sales,  
(sales - profit) AS raw_cost  
FROM samplesuperstore ORDER BY raw_cost DESC;
```

OUTPUT:-

ship mode	sub-category	profit	sales	rawcost
Standard Class	Machines	-1811.0784	22638.48	24449.558399999998
Same Day	Machines	-3839.9904	7999.98	11839.9704
Standard Class	Machines	-6599.978	4499.985	11099.963
Standard Class	Machines	-1359.992	8159.952	9519.944
Standard Class	Copiers	8399.976	17499.95	9099.974
Second Class	Supplies	327.506	8187.65	7860.143999999999
First Class	Copiers	3919.9888	11199.968	7279.979200000001
Standard Class	Copiers	1119.9968	8399.976	7279.979200000001
First Class	Copiers	6719.9808	13999.96	7279.979199999999
Standard Class	Machines	2365.9818	9099.93	6733.948200000001
Second Class	Tables	-1862.3124	4297.644	6159.9564

- Write a query to display the ship mode, sub-category, profit, sales, margin percentage?

QUERY :-

```
SELECT `ship mode`, `sub-category`,  
       profit, sales,  
       ROUND((profit / sales) * 100, 1) As margin_percentage  
FROM samplesuperstore  
ORDER BY margin_percentage DESC;
```

OUTPUT :-

ship mode	sub-category	profit	sales	margin_percentage
Same Day	Paper	24.47	48.94	50
Same Day	Fasteners	8.95	17.9	50
Standard Class	Binders	635.495	1270.99	50
Standard Class	Binders	6.15	12.3	50
Standard Class	Binders	4946.37	9892.74	50
Second Class	Labels	22.2	44.4	50
Standard Class	Fasteners	18.24	36.48	50
Same Day	Art	3.36	6.72	50
Second Class	Paper	17.22	34.44	50
Same Day	Binders	19.08	38.16	50
Standard Class	Furnishings	297.69	595.38	50
Standard Class	Envelopes	35.985	71.97	50

- Write a query to display the which cities getting highest margin percentage in united states ?

QUERY :-

```
SELECT `ship mode`,country,city,`sub-category`,profit,sales,  
       ROUND((profit / sales) * 100, 1) As margin_percentage  
FROM samplesuperstore  
WHERE country='UNITED STATES'  
ORDER BY margin_percentage DESC ;
```

OUTPUT :-

ship mode	country	city	sub-category	profit	sales	margin_percentage
Same Day	United States	Los Angeles	Art	3.36	6.72	50
Second Class	United States	New York City	Paper	17.22	34.44	50
Same Day	United States	Oklahoma City	Binders	19.08	38.16	50
Standard Class	United States	Los Angeles	Furnishings	297.69	595.38	50
Standard Class	United States	San Clemente	Envelopes	35.985	71.97	50
Second Class	United States	Richmond	Binders	8.88	17.76	50
Standard Class	United States	San Francisco	Paper	25.02	50.04	50
Standard Class	United States	Long Beach	Labels	14.8	29.6	50
Standard Class	United States	Los Angeles	Envelopes	5.58	11.16	50
First Class	United States	New York City	Envelopes	11.995	23.99	50
Second Class	United States	Indianapolis	Binders	8.69	17.38	50
First Class	United States	Midland	Binders	1906....	3812.97	50

- Write a query to display the which cities getting highest margin percentage in united states with starting city with alphabet 's' ?

QUERY :-

```
SELECT `ship mode`,country,city,`sub-category`, profit, sales,  
ROUND((profit / sales) * 100, 1) AS margin_percentage  
FROM samplesuperstore  
WHERE country='UNITED STATES' AND CITY LIKE 'S%'  
ORDER BY margin_percentage DESC ;
```

OUTPUT :-

ship mode	country	city	sub-category	profit	sales	margin_percentage
First Class	United States	San Francisco	Paper	25.48	50.96	50
First Class	United States	San Francisco	Envelopes	10.94	21.88	50
Standard Class	United States	San Francisco	Paper	3.71	7.42	50
Standard Class	United States	Sioux Falls	Binders	30.415	60.83	50
First Class	United States	Springfield	Envelopes	16.98	33.96	50
Second Class	United States	Summerville	Phones	39.55	79.1	50
Standard Class	United States	Springfield	Binders	116.48	232.96	50
Standard Class	United States	San Francisco	Fasteners	15.2	30.4	50
Standard Class	United States	San Francisco	Paper	97.88	195.76	50
Standard Class	United States	San Francisco	Envelopes	11.58	23.16	50
Standard Class	United States	Seattle	Labels	22.2	44.4	50

- Write a query to display the top performing cities in that region ?

QUERY :-

```
with RegionalSales as ( select city,Region,`Sub-Category`, SUM(Sales) AS  
Total_Sales  
FROM samplesuperstore  
GROUP BY city,Region, `Sub-Category`),  
RankedSales AS ( SELECT *, RANK() OVER (PARTITION BY Region ORDER  
BY Total_Sales DESC) as Rank_In_Region  
FROM RegionalSales)  
SELECT * FROM RankedSales  
WHERE Rank_In_Region = 1;
```

OUTPUT :-

city	Region	Sub-Category	Total_Sales	Rank_In_Region
Lafayette	Central	Copiers	17499.95	1
New York City	East	Chairs	40149.693000000001	1
Jacksonville	South	Machines	24155.48	1
Los Angeles	West	Phones	29503.031999999992	1

- write a query to display the top 3 profitable branches in united states?

QUERY :-

```
SELECT `ship mode`, country, `POSTAL CODE`,  
city, `sub-category`, profit  
FROM samplesuperstore  
WHERE country='UNITED STATES'  
ORDER BY PROFIT DESC  
LIMIT 3;
```

OUTPUT :-

ship mode	country	POSTAL CODE	city	sub-category	profit
Standard Class	United States	47905	Lafayette	Copiers	8399.976
First Class	United States	98115	Seattle	Copiers	6719.9808
Standard Class	United States	19711	Newark	Copiers	5039.9856

- write a query to display the top profitable cities in each category ?

QUERY :-

```
with temp as(SELECT `ship mode`,country,`postal code`,city,`sub-  
category`, profit, category,sum(Sales) AS Total_Sales  
FROM samplesuperstore  
group by `ship mode`,country,`postal code`,city,`sub-category`,  
profit, category),  
temp2 as(select*,rank() over( partition by category order by  
total_sales ) as ranks  
from temp)  
select* from temp2  
where ranks = 1;
```

OUTPUT :-

ship mode	country	POSTAL CODE	city	sub-category	profit	category	Total_Sales	ranks
Standard Class	United States	60653	Chicago	Furnishings	-0.9933	Furniture	1.892	1
Standard Class	United States	77095	Houston	Appliances	-1.11	Office Supplies	0.444	1
First Class	United States	94110	San Francisco	Accessories	0.4356	Technology	0.99	1

- write a query to display the top loser's from their respective region ?

QUERY :-

```
with RegionalSales as (select city,Region,`postal code`, `sub-category`,  
sum(profit) as Total_Sales  
FROM samplesuperstore  
group by city,Region, `sub-category`, `postal code`),  
RankedSales as  
( select *,  
rank() over(partition by Region order by Total_Sales asc) as Rank_In_Region  
from RegionalSales)  
select * from RankedSales  
Where Rank_In_Region =1;
```

OUTPUT :-

city	Region	postal code	Sub-Category	Total_Sales	Rank_In_Region
Chicago	Central	60653	Binders	-4290.7478	1
Lancaster	East	43130	Machines	-6599.978	1
Burlington	South	27217	Machines	-3839.9904	1
Louisville	West	80027	Machines	-3399.98	1

- write a query to display the top sales sub-category by quantity?

QUERY :-

```
SELECT region,category,'sub-category',sales,quantity  
FROM samplesuperstore  
order by Quantity desc  
limit 5;
```

OUTPUT :-

region	category	sub-category	sales	quantity
West	Technology	Accessories	223.58	14
East	Furniture	Furnishings	54.992	14
East	Technology	Accessories	1119.888	14
East	Furniture	Chairs	427.644	14
Central	Office Supplies	Art	54.208	14

- write a query to display the top 5 individual orders with the highest quantity of items sold in the South Region?

QUERY :-

```
SELECT region,category,`sub-category`,sales,quantity
FROM samplesuperstore
WHERE REGION = 'SOUTH'
order by Quantity DESC
limit 5;
```

OUTPUT :-

region	category	sub-category	sales	quantity
South	Office Supplies	Paper	295.456	14
South	Furniture	Chairs	1819.86	14
South	Office Supplies	Paper	319.76	14
South	Furniture	Furnishings	192.22	14
South	Technology	Accessories	279.86	14

- write a query to display the unit price of top 5 best performed branches ?

QUERY :-

```
SELECT region,category,'sub-category',  
sales,quantity,(sales / quantity) as unit_price  
FROM samplesuperstore  
order by Quantity DESC  
limit 5;
```

OUTPUT :-

region	category	sub-category	sales	quantity	unit_price
West	Technology	Accessories	223.58	14	15.97
East	Furniture	Furnishings	54.992	14	3.928
East	Technology	Accessories	1119.888	14	79.99199999999999
East	Furniture	Chairs	427.644	14	30.546
Central	Office Supplies	Art	54.208	14	3.872

- write a query to display the top 5 sold sub category with their unit price of the product that are ship mode by first class ?

QUERY :-

```
SELECT `ship mode`,region,category,  
`sub-category`,sales,quantity,(sales / quantity) as unit_price  
FROM samplesuperstore  
where `ship mode` ='first class'  
order by Quantity desc  
limit 5;
```

OUTPUT :-

ship mode	region	category	sub-category	sales	quantity	unit_price
First Class	West	Office Supplies	Binders	38.388	14	2.742
First Class	West	Office Supplies	Binders	61.776	13	4.752000000000001
First Class	South	Technology	Accessories	272.61	13	20.970000000000002
First Class	East	Furniture	Furnishings	547.3	13	42.099999999999994
First Class	West	Furniture	Tables	1673.184	12	139.432

- write a query to display the details of top 3 highest order sales valued branches with unit price and also check weather they are in losses or profits?

QUERY :-

```
select `ship mode`,country,city,state,`postal code`,region,category,`sub-  
category`,sales,quantity,profit,(sales / quantity) as unit_price ,case when  
(profit / quantity) > 0 then 'profit' else 'loss' end as unit_analysis  
from samplesuperstore  
order by sales desc  
limit 3;
```

OUTPUT:-

ship mode	country	city	state	postal code	region	category	sub-category	sales	quantity	profit	unit_price	unit_analysis
Standard Class	United States	Jacksonville	Florida	32216	South	Technology	Machines	22638.48	6	-1811.0784	3773.08	loss
Standard Class	United States	Lafayette	Indiana	47905	Central	Technology	Copiers	17499.95	5	8399.976	3499.9900000000002	profit
First Class	United States	Seattle	Washington	98115	West	Technology	Copiers	13999.96	4	6719.9808	3499.99	profit

- write a query to display the top 5 branches who are offering highest average discount among all the branches ?

QUERY :-

```
select country,city,'Postal Code',avg(Discount) as  
avg_discount  
from samplesuperstore  
group by country,city,'Postal Code'  
order by avg_discount desc  
limit 5;
```

OUTPUT :-

country	city	Postal Code	avg_discount
United States	Danville	61832	0.8
United States	Abilene	79605	0.8
United States	Missouri City	77489	0.8
United States	Deer Park	77536	0.8
United States	Romeoville	60441	0.8

- write a query to display the details of cities those are having above average profits?

QUERY :-

```
select state,city,'postal code',sum(profit) as total_profit
from samplesuperstore
group by state,city,'postal code'
having sum(profit) > (select avg(Profit) from samplesuperstore);
```

OUTPUT :-

state	city	postal code	total_profit
Kentucky	Henderson	42420	4518.8169
California	Los Angeles	90036	4452.9468000000001
California	Los Angeles	90032	4149.2141999999998
Washington	Seattle	98103	7117.6170000000001
Texas	Fort Worth	76106	298.7327
Wisconsin	Madison	53711	1123.65520000000004
Utah	West Jordan	84084	102.6723
California	San Francisco	94109	4397.4533000000001
Nebraska	Fremont	68025	179.4696
Utah	Orem	84057	514.6335
California	Los Angeles	90049	7793.1144999999997
Minnesota	Eagan	55122	87.0338

- write a query to display the unique categories and also count of respective unique sub categories?

QUERY :-

```
select category,count(distinct category) as  
total_unique_categories, count(distinct `sub-category`) as  
total_unique_subcategories  
from samplesuperstore  
group by category;
```

OUTPUT :-

category	total_unique_categories	total_unique_subcategories
Furniture	1	4
Office Supplies	1	9
Technology	1	4

- write a query to display the unique ship mode and how many orders we get from each?

QUERY :-

```
select distinct `ship mode`,  
count(`ship mode`) over (partition by `ship mode`) as  
total_no_of_shipment  
from samplesuperstore;
```

OUTPUT :-

ship mode	total_no_of_shipment
First Class	1538
Same Day	543
Second Class	1945
Standard Class	5968

- write a to display the count of stores in state wise?

QUERY :-

```
select distinct state,count(city) over(partition by state) as  
no_of_branches  
from samplesuperstore ;
```

OUTPUT :-

state	no_of_branches
Alabama	61
Arizona	224
Arkansas	60
California	2001
Colorado	182
Connecticut	82
Delaware	96
District of Columbia	10
Florida	383
Georgia	184
Idaho	21
Illinois	492
Indiana	149
Iowa	30
Kansas	24
Kentucky	139
Louisiana	42

- write a query to display the whose store sold quantity above average quantity?

QUERY :-

```
select city, `postal code`, avg(quantity) as above_avg  
from samplesuperstore  
group by city, `postal code`  
having avg(quantity)>(select avg(quantity)  
                        from samplesuperstore);
```

OUTPUT :-

city	postal code	above_avg
Henderson	42420	3.8542
Los Angeles	90036	3.8361
Fort Lauderdale	33311	4.3333
Los Angeles	90032	4.0556
Concord	28027	5.5000
Fremont	68025	4.1250
Los Angeles	90049	3.8411
Richardson	75080	6.0000
Naperville	60540	4.1250
Troy	12180	3.8000
Los Angeles	90004	3.9316
Gilbert	85234	3.8667
Springfield	22153	3.9600
Memphis	38109	3.8667

- write a to display the count of stores in state wise whose state name starts with 'g' alphabet?

QUERY :-

```
select distinct state,  
               count(city) over(partition by state) as no_of_branches  
from samplesuperstore  
where state like 'g%';
```

OUTPUT :-

state	no_of_branches
Georgia	184

- write a query to display the details of top 3 state that having lowest branches ?

QUERY :-

```
SELECT t1.State, t1.City, t1.`Postal Code`, t1.Sales,  
t1.Profit,t2.Total_branches  
FROM samplesuperstore t1  
JOIN (SELECT State, COUNT(*) as Total_branches  
FROM samplesuperstore  
GROUP BY State  
ORDER BY COUNT(*) ASC  
LIMIT 3) t2 ON t1.State = t2.State  
ORDER BY t2.Total_branches ASC, t1.State;
```

OUTPUT :-

State	City	Postal Code	Sales	Profit	Total_branches
Wyoming	Cheyenne	82001	1603.136	100.196	1
West Virginia	Wheeling	26003	82.4	40.376	4
West Virginia	Wheeling	26003	6.24	3.0576	4
West Virginia	Wheeling	26003	447.84	219.4416	4
West Virginia	Wheeling	26003	673.344	-76.9536	4
North Dakota	Fargo	58103	131.94	35.6238	7
North Dakota	Fargo	58103	25.92	8.2944	7
North Dakota	Fargo	58103	704.76	162.0948	7
North Dakota	Fargo	58103	7.41	3.4827	7
North Dakota	Fargo	58103	21.5	7.095	7
North Dakota	Fargo	58103	2.48	0.868	7
North Dakota	Fargo	58103	25.9	12.691	7

- write query to display the states and count of branches who are using ship mode 'first class'?

QUERY :-

```
select distinct state, `ship mode` count(city)
over(partition by state) as no_of_branches
from samplesuperstore
where `ship mode` = 'first class'
order by no_of_branches ;
```

OUTPUT :-

state	ship mode	no_of_branches
New Mexico	First Class	1
Iowa	First Class	1
Montana	First Class	1
New Hampshire	First Class	2
South Dakota	First Class	2
Mississippi	First Class	3
South Carolina	First Class	3
Idaho	First Class	4
Utah	First Class	4
Oklahoma	First Class	5
Nevada	First Class	5
Kansas	First Class	6
Nebraska	First Class	7
Missouri	First Class	8
Alabama	First Class	9

- write a query to check whether there is any store having postal code number '19140'?

QUERY :-

```
select *  
from samplesuperstore  
where `postal code`=19140  
limit 1;
```

OUTPUT :-

Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
Second Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furniture	Chairs	71.372	2	0.3	-1.0196

- write a query to display the consumer segment branches without duplicate rows and sort in ship mode?

QUERY :-

```
with temp as(select `ship  
mode`,`segment`,`region`,`country`,`city`,`state`,`postal code`  
from samplesuperstore  
where segment='consumer')  
select distinct* from temp  
order by `ship mode`;
```

OUTPUT :-

ship mode	segment	region	country	city	state	postal code
First Class	Consumer	Central	United States	Abilene	Texas	79605
First Class	Consumer	Central	United States	Allen	Texas	75002
First Class	Consumer	Central	United States	Arlington	Texas	76017
First Class	Consumer	Central	United States	Aurora	Illinois	60505
First Class	Consumer	Central	United States	Austin	Texas	78745
First Class	Consumer	Central	United States	Cedar Rapids	Iowa	52402
First Class	Consumer	Central	United States	Chicago	Illinois	60610
First Class	Consumer	Central	United States	Chicago	Illinois	60623
First Class	Consumer	Central	United States	Chicago	Illinois	60653
First Class	Consumer	Central	United States	Columbus	Indiana	47201
First Class	Consumer	Central	United States	Dallas	Texas	75081
First Class	Consumer	Central	United States	Dallas	Texas	75220
First Class	Consumer	Central	United States	Decatur	Illinois	62521
First Class	Consumer	Central	United States	Des Plaines	Illinois	60016

- write a query to find the average revenue generating by New York state?

QUERY :-

```
select avg(sales) as `average revenue ($)`  
from Samplesuperstore  
where state='New York';
```

OUTPUT :-

average revenue (\$)
275.5995310283686